What is a tensor?

In deep learning, think of a tensor as an n-dimensional array of numbers.



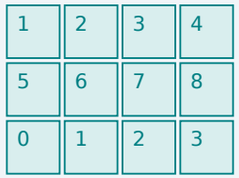
0D / SCALAR

One number



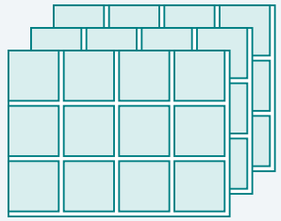
1D / VECTOR

One axis



2D / MATRIX

Rows + columns



3D / TENSOR

A stack of tables

Axes	Temperature example	Shape
0	One measurement	()
1	24 hourly measurements	(24,)
2	10 cities, 24 hours	(10, 24)
3	7 days, 10 cities, 24 hours	(7, 10, 24)

Dimensions are axes.

An axis tells you how to locate a value. Shape tells you how many positions exist along each axis.

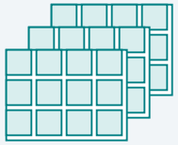
7 x 10 x 24 = 1,680 values

Multiply the axis sizes to count entries.
A matrix is specifically 2D; “n-dimensional array” is the general description.

Tensors you can picture

First explain how the values are organized. Then explain the work needed to process them.

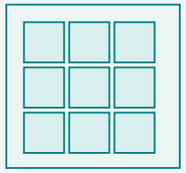
01 Spreadsheet collection



Rows x columns x months

More sheets mean more cells to calculate.

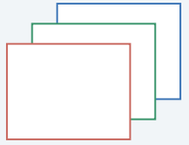
02 Apartment building



Floors x apartments x rooms

More positions mean more values to process.

03 Color photograph



R / G / B

Height x width x 3 channels

Double height AND width: 4x the values.

04 Video flipbook



Frames x height x width x channels

100 frames contain 100x the values of one frame.

05 Restaurant order book



Orders x dishes x ingredients

Many similar calculations can run in parallel.

More axes do not automatically mean more work.

5D: (2, 2, 2, 2, 2)
32 values

2D: (1,000, 1,000)
1,000,000 values

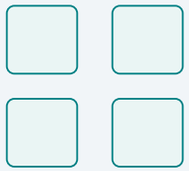
03 / COMPUTING THE NUMBERS

CPU, GPU and TPU

Different strengths, not a simple ranking. Sketches show the idea, not literal core counts.

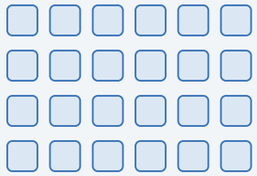
CPU

Versatile workers



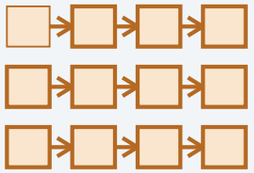
GPU

A parallel team



TPU

A specialized factory



	CPU	GPU	TPU
Full name	Central Processing Unit	Graphics Processing Unit	Tensor Processing Unit
Strength	General tasks, logic and control flow	Many similar calculations in parallel	Specialized large matrix operations
Classroom use	Python, pandas, conventional ML	Neural-network training; suitable pretrained models	Notebooks designed for TPUs
Watch out	Large neural networks may be slow	Code must actually use the GPU	Needs compatible tooling and a suitable workload

Why deep learning needs compute

Large tensors pass through many layers. Training also computes gradients and updates weights, repeatedly across batches and epochs. GPUs and TPUs can accelerate suitable operations.

Which resource should I use?

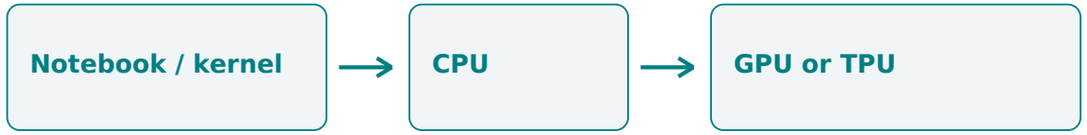
Start with the task. Choose an accelerator only when the notebook can use it.

Your task	Start with	Reason
Python, pandas, plots, ordinary scikit-learn	CPU	General work usually needs no accelerator.
Neural network in a GPU-compatible notebook	GPU	Parallel tensor operations can reduce runtime.
Tutorial explicitly written for a TPU	TPU	The code is prepared for TPU execution.
Call a remote AI model API	CPU	The provider runs the model on its servers.

COLAB SETUP

Runtime > Change runtime type > Hardware accelerator
Availability and usage limits vary. Selecting an accelerator does not automatically make your code use it.

The CPU remains part of the system.



Three myths to challenge

“GPU speeds up everything.” Small jobs and data transfers may erase the gain.
“TPU is always better.” Performance depends on the workload.
“Better hardware means a better model.” It does not repair poor data or evaluation.

Official sources / further reading
Google Colab FAQ
Google Cloud: Introduction to TPU
Colab options may change.